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College of Science and Technology
Rinchending: Bhutan

WEB101
WEB APPLICATION FUNDAMEN
(SS2026)

Practical 2 Report

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Programme: BE 1SWE
Date: 05.04.2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

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ACKNOWLEDGEMENT

First of all, I would like to thank the College of Science and Technology for giving me the important required resources for me to be able to do this practical. I am also very much thankful to our WEB module teacher, Mr. Prakash Chhetri, for helping us with his guidance and support while doing the assignment, where it helped me to understand better about the basics of web application development, not just rely on ai.

AIM AND OBJECTIVES

Aim

Is to make a simple weather web application by using a real REST API to retrieve original weather information or data of any city in the world along with the ability to let the user save,edit and delete any locations thought HTTP methods.

Objectives

- Learn the functions of REST API and how it works

- Create a account in OpenWeatheMap API to get real weather information

- For practice of POST, PUT and DELETE requests use JSON place holder API

- Make a simple working weather dashboard interface with use of HTML,CSS and JavaScript

- A clean way to show the API response data on the screen for the user perspective

- Like always, have the standard proper error management just in case errors pop up.

Background Information

These days, new web applications are made using different frameworks and tools to make the process faster, cleaner, and easier to handle over long durations, whereas in this particular assignment, I have made use of REST API where the API stands for Application Programming Interface which is nothing but the means of communication between two programs on the net. When my app is in need of data from another server ,the APIs sends an immediate request to the server and back to it.

REST API

The full form of REST is Representational State Transfer which is nothing but the way of making APIs by using means like the HTTP requests where most of websites or apps that we use in our daily life puts use of REST APIs to retrive their information such as when we open our phone to check the weather , the app sends a Get request to the weather API just like what I build in this practical.

HTTP Methods

The Four main HTTP methods in used in the practical are the following:

GET – to retrieve the asked data fetching weather from the OpenWeatherMap API
POST – to make a new data into the server where I used it to save any new locations into the JSON Place Holder API
PUT – Just to edit any already existing data like in the saved locations section.
DELETE – Just to delete any unnecessary or no longer required data by deleting it in the saved list

Fetch API

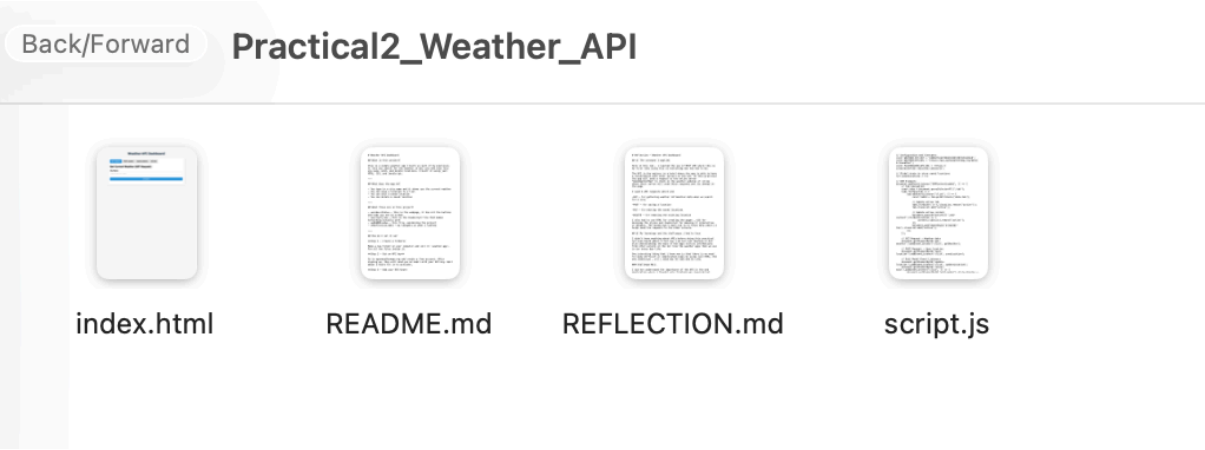
This is nothing but a already existing built-in JavaScript tool that was already there which lets one send HTTP requests to the net. Instead of needing to download many libraries, with just this tool alone I can use fetch() in JavaScript where it will send the request for my sake.

JSON

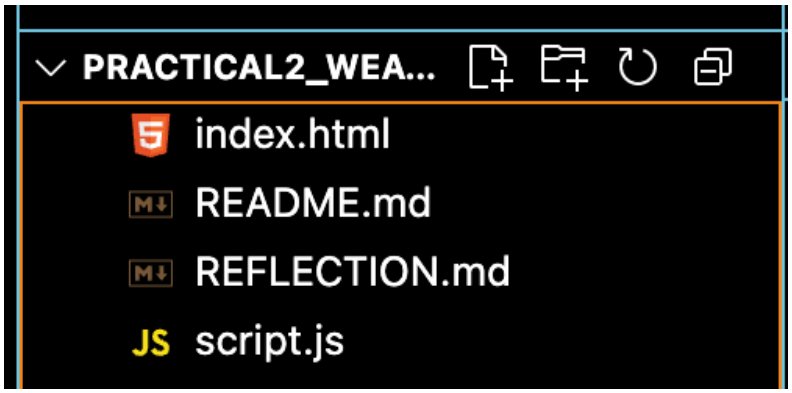
JavaScript Object Notation the full form of JSON is just a standard format which APIs use to send requests and responses of data between two programs. Example- {"city": "London", "temp": 15} where use of JSON.stringify() to send data and JSON.parse() to read datas from the server

Implementation/ PROCEDURE

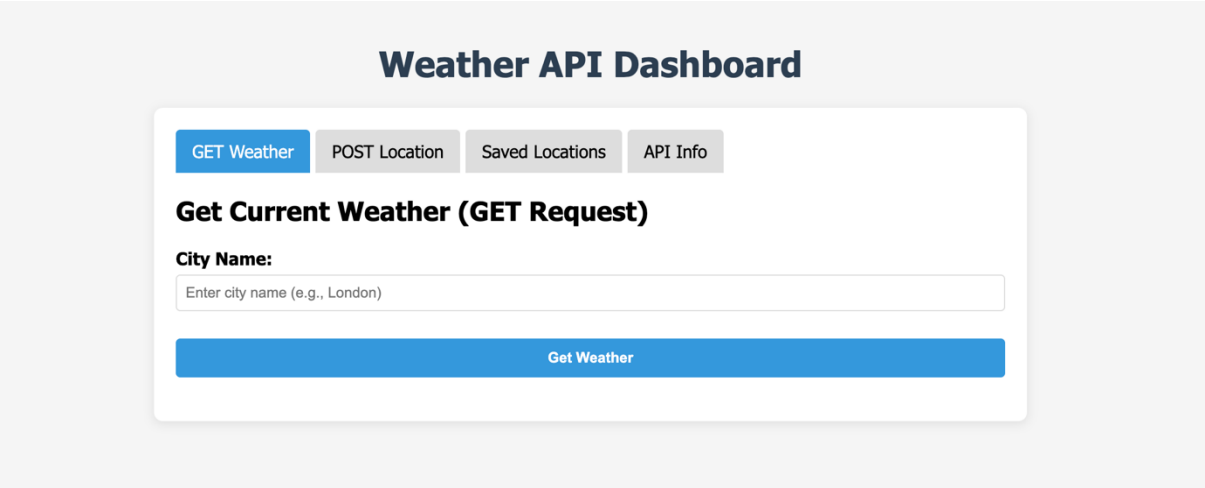
Step 1: Create a new folder called weather-app with two files index.html and script.js



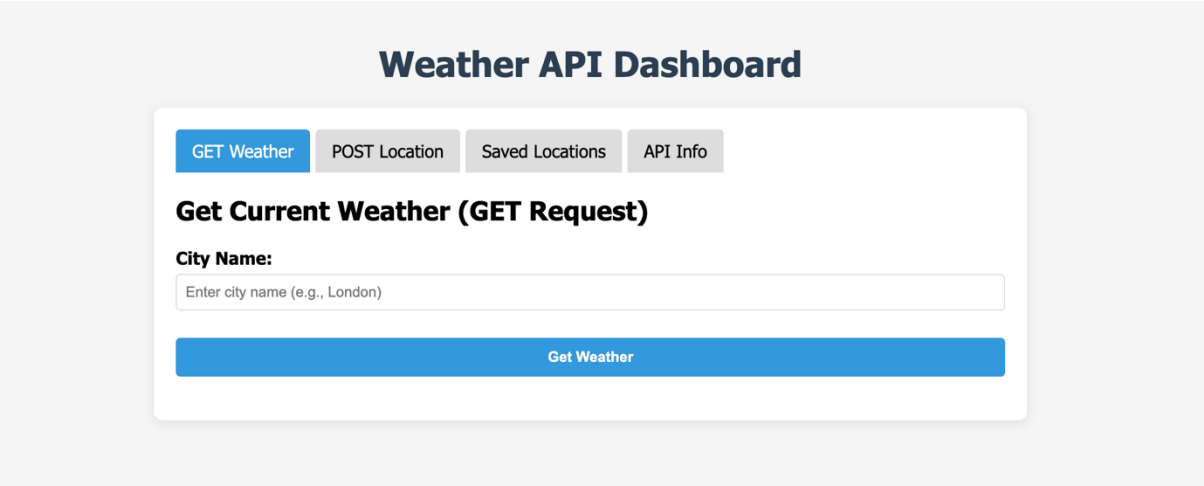
Step 2: Open the project folder in VS Code.

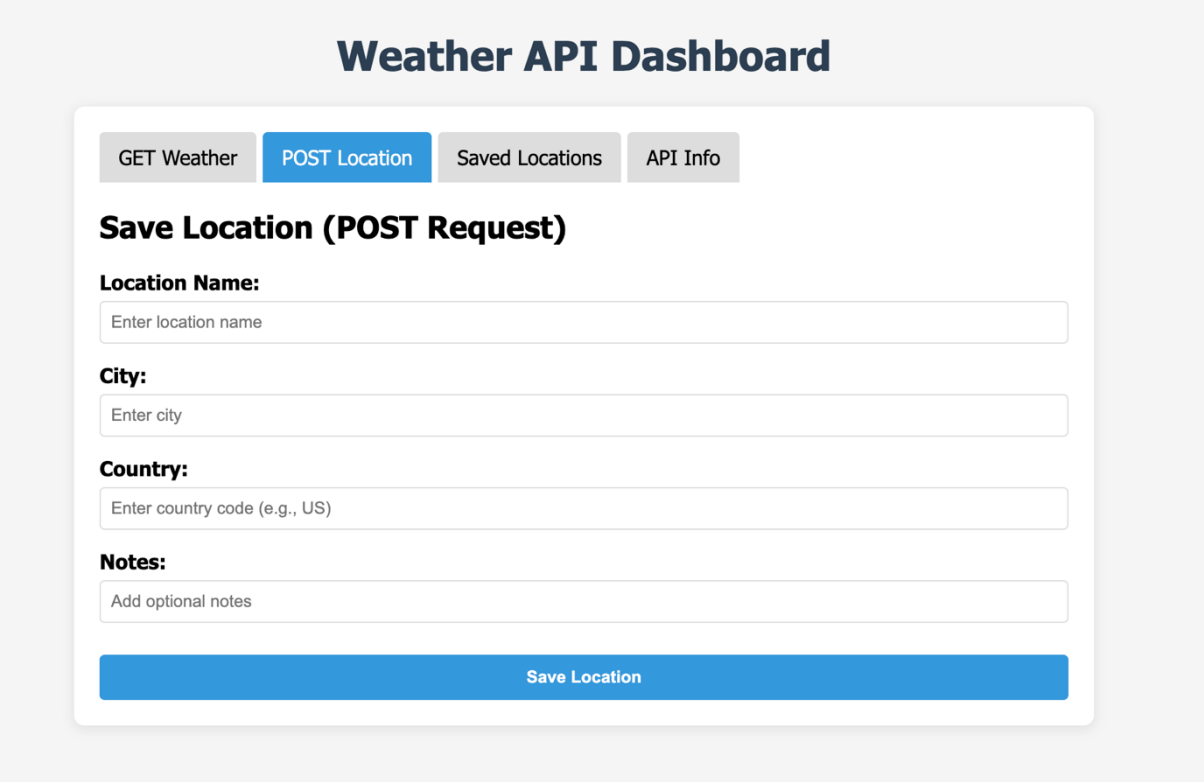


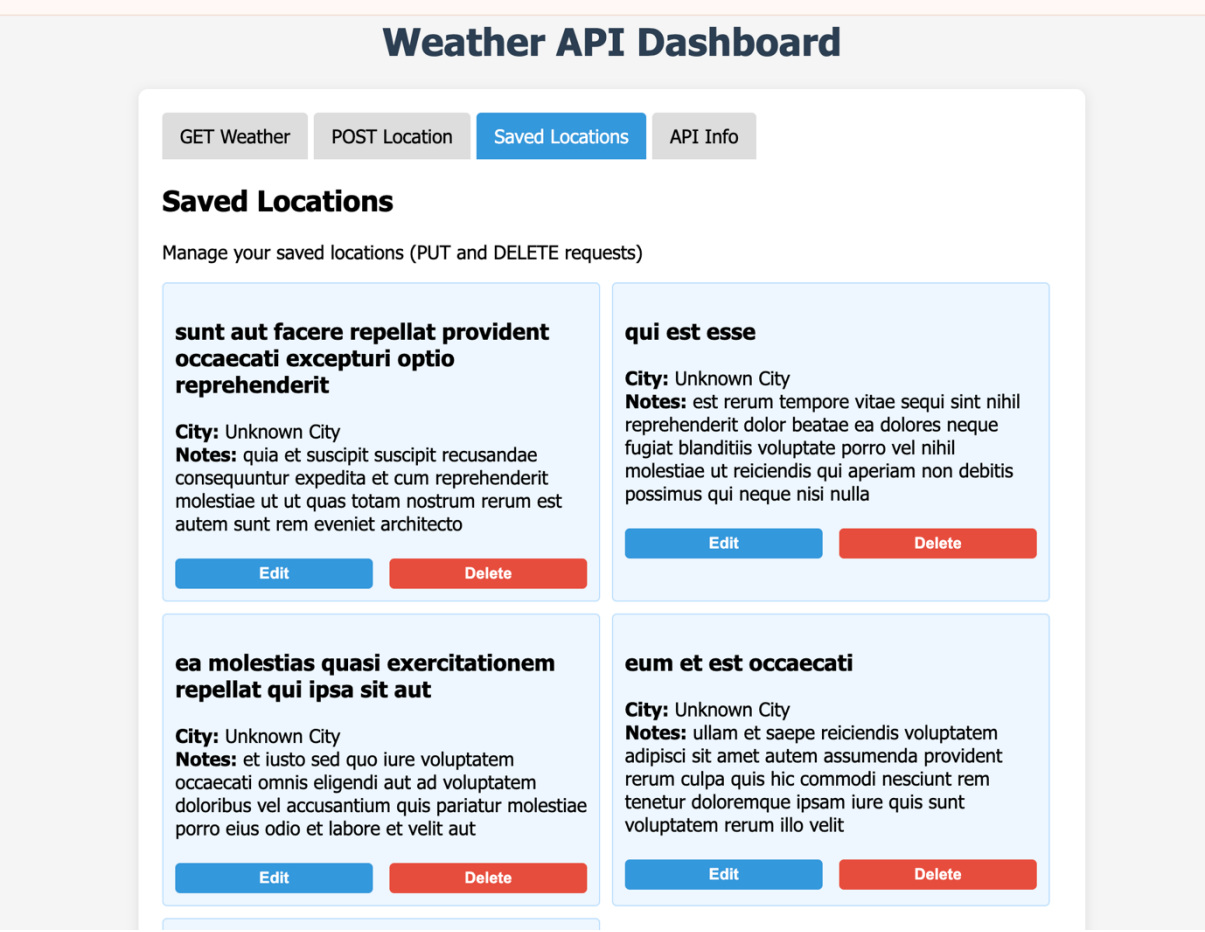
Step 3: HTML structure with four tabs displayed in the browser.



Step 4: JavaScript functionality with working tabs and buttons.






Weather API Dashboard
GET Weather
POST Location
Saved Locations
API Info
Saved Locations
Manage your saved locations (PUT and DELETE requests)
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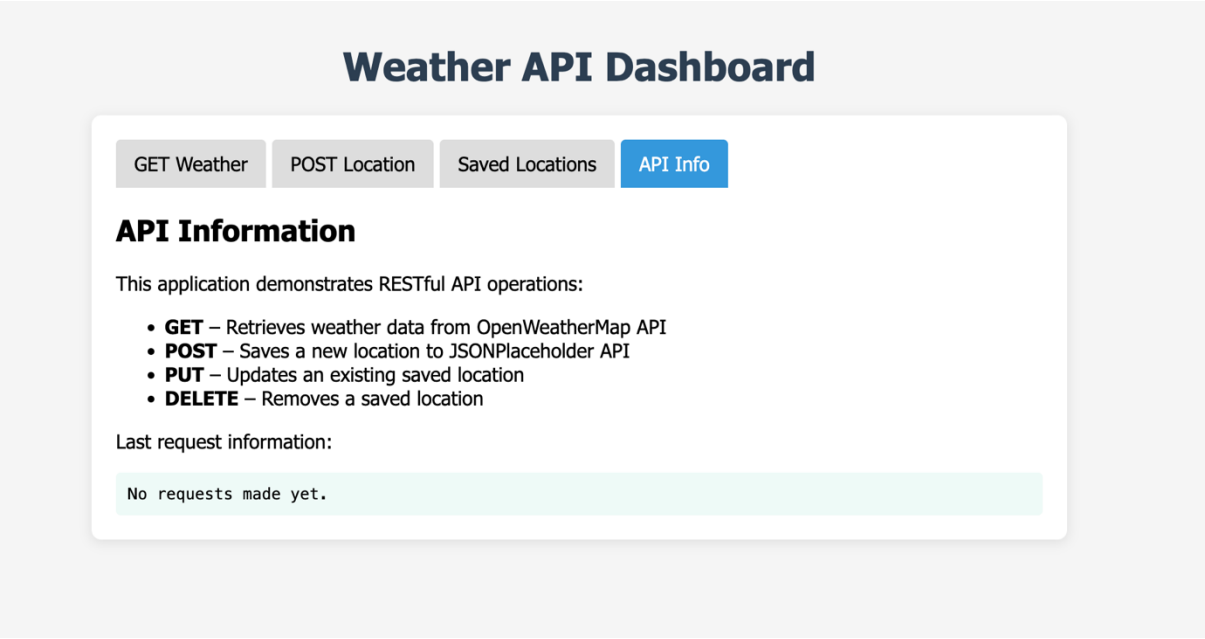
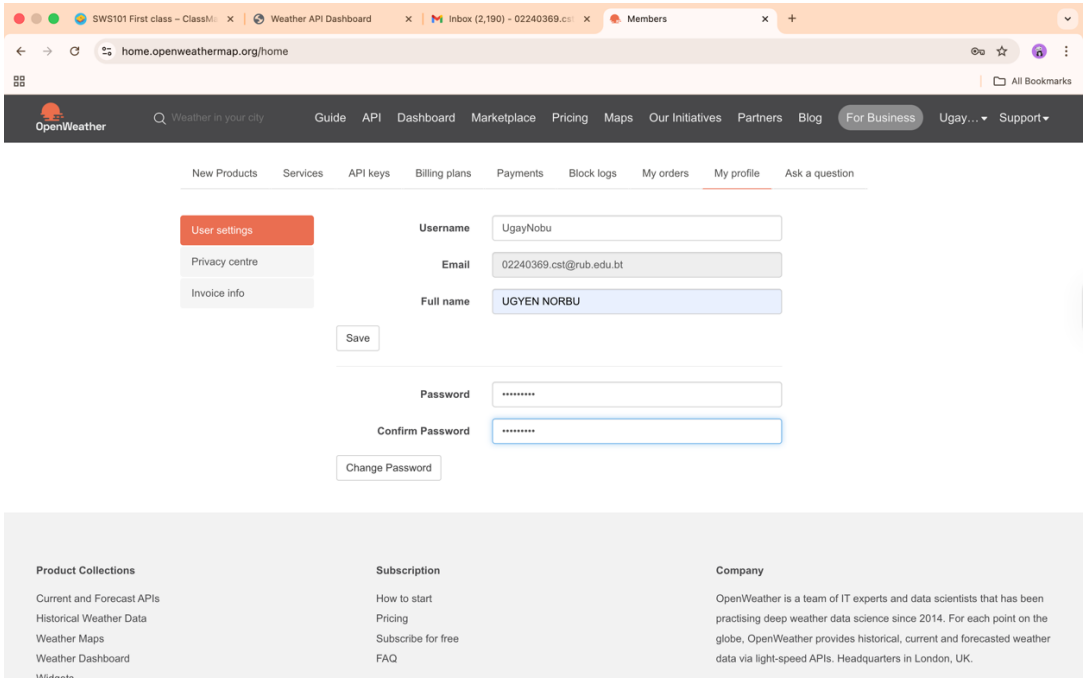

Weather API Dashboard
GET Weather
POST Location
Saved Locations
API Info
API Information
This application demonstrates RESTful API operations:
GET – Retrieves weather data from OpenWeatherMap API
POST – Saves a new location to JSONPlaceholder API
PUT – Updates an existing saved location
DELETE – Removes a saved location
Last request information:
No requests made yet.

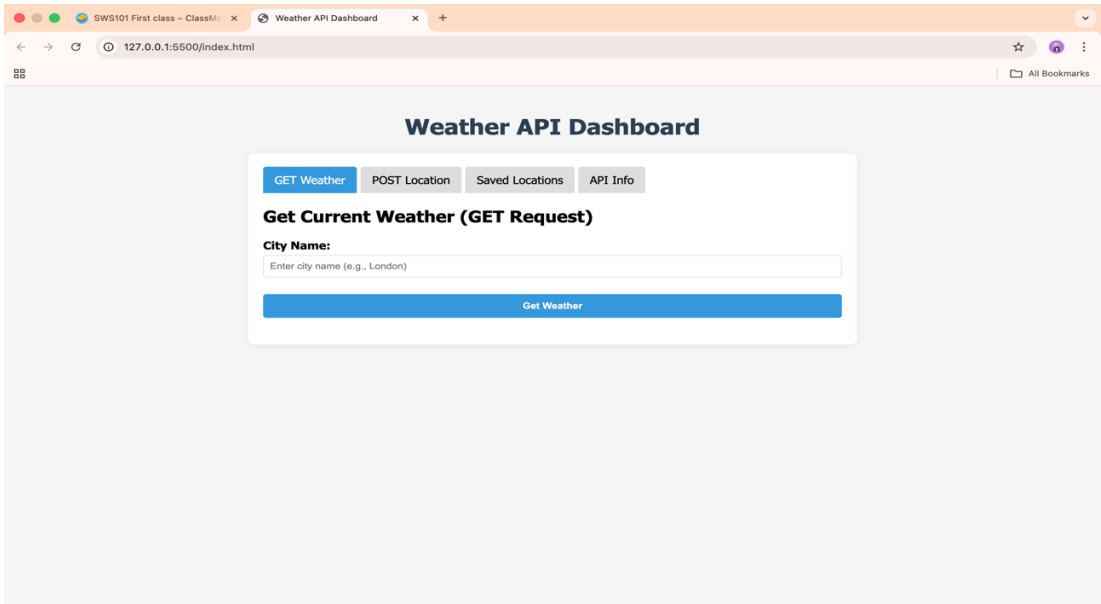
Step 5: Sign up for the weather website at openweathermap.org



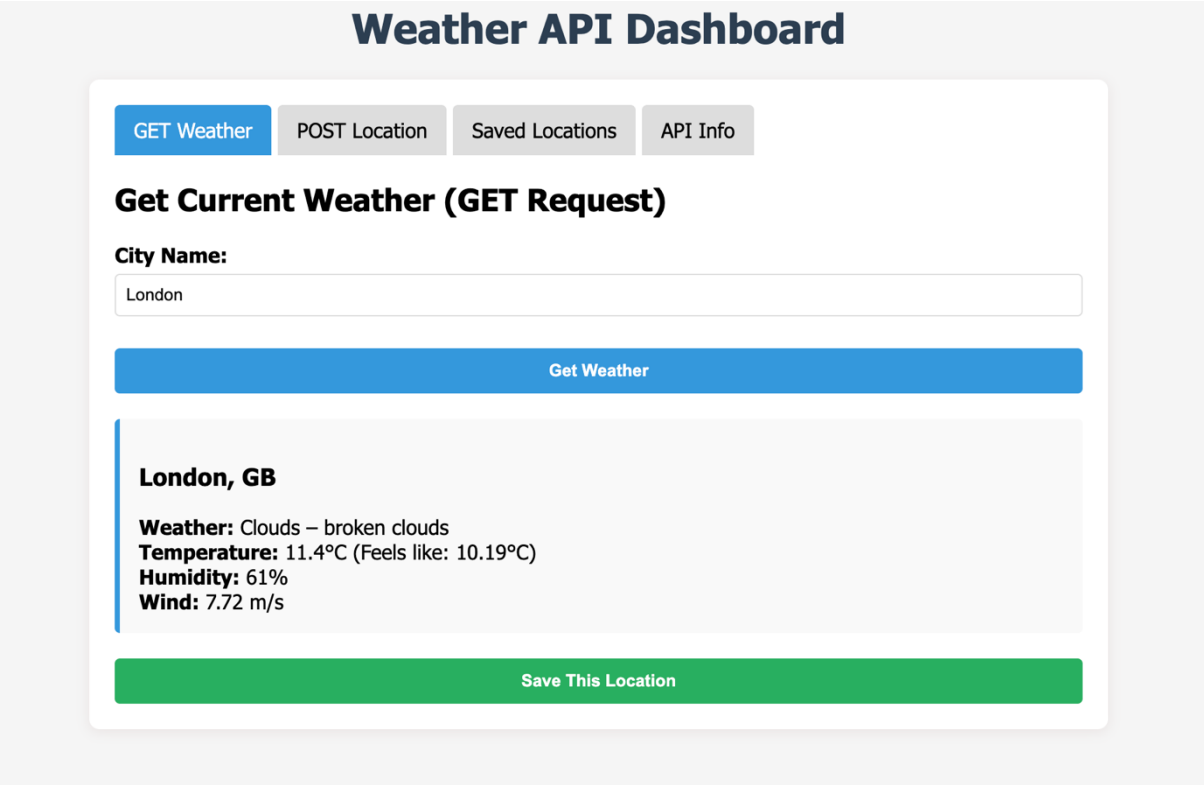
Step 6: API key added to line 2 of script.js

```
1  // Configuration and Constants        You, 2 weeks ago • Practical2_Weather_API
2  const WEATHER_API_KEY = '6488e79cabf4b8dcb9912091524a18e0';
3  const WEATHER_API_URL = 'https://api.openweathermap.org/data/2.5/weather';
4  const PLACEHOLDER_API_URL = 'https://jsonplaceholder.typicode.com/posts';
```

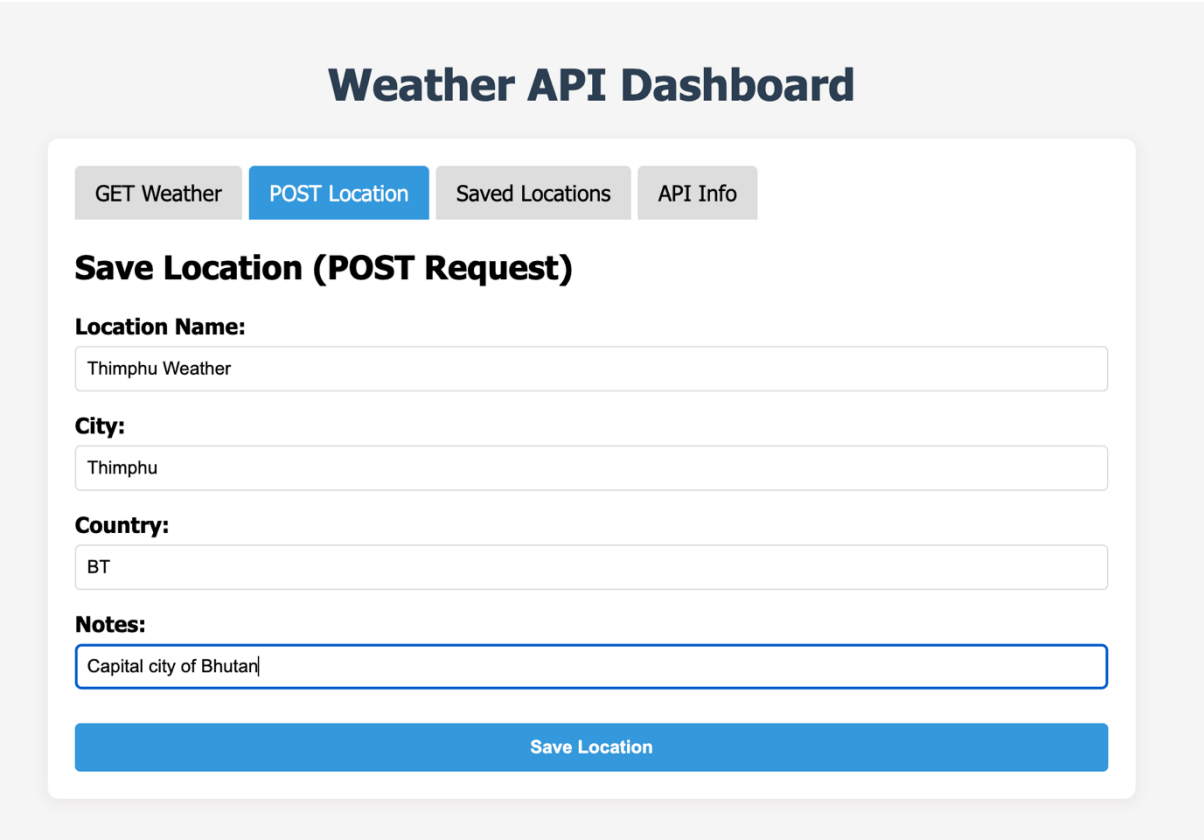
Step 7: Open the app using the Go Live button in VS Code.

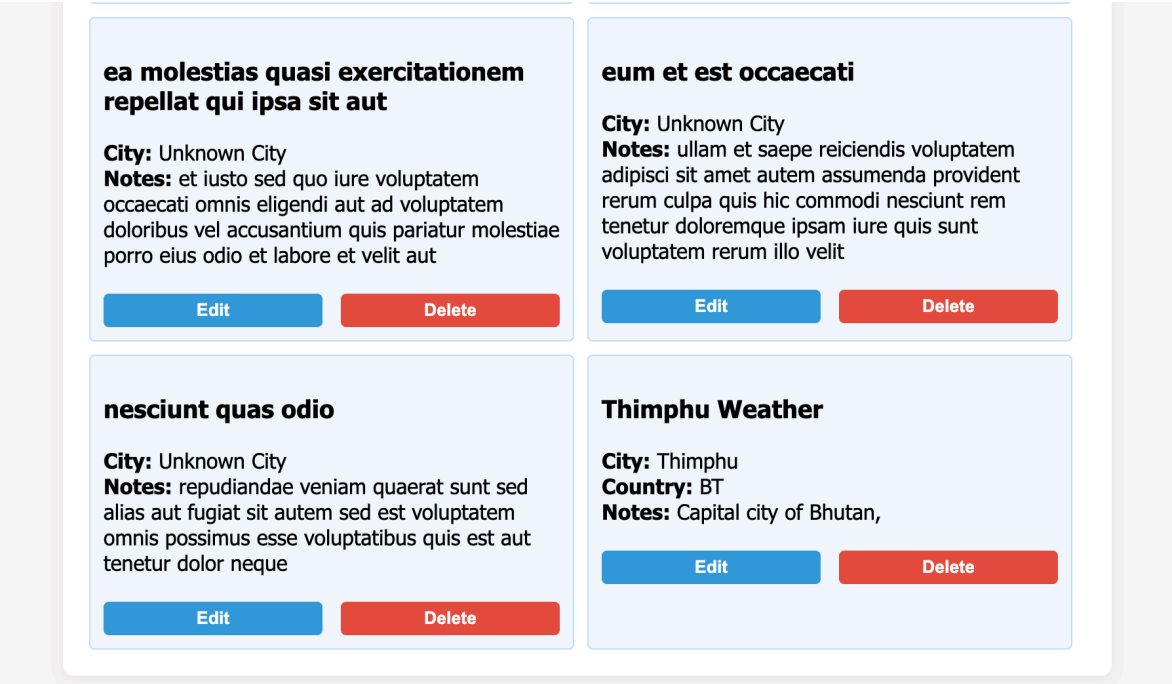


Step 8: GET request tested by searching for a city.

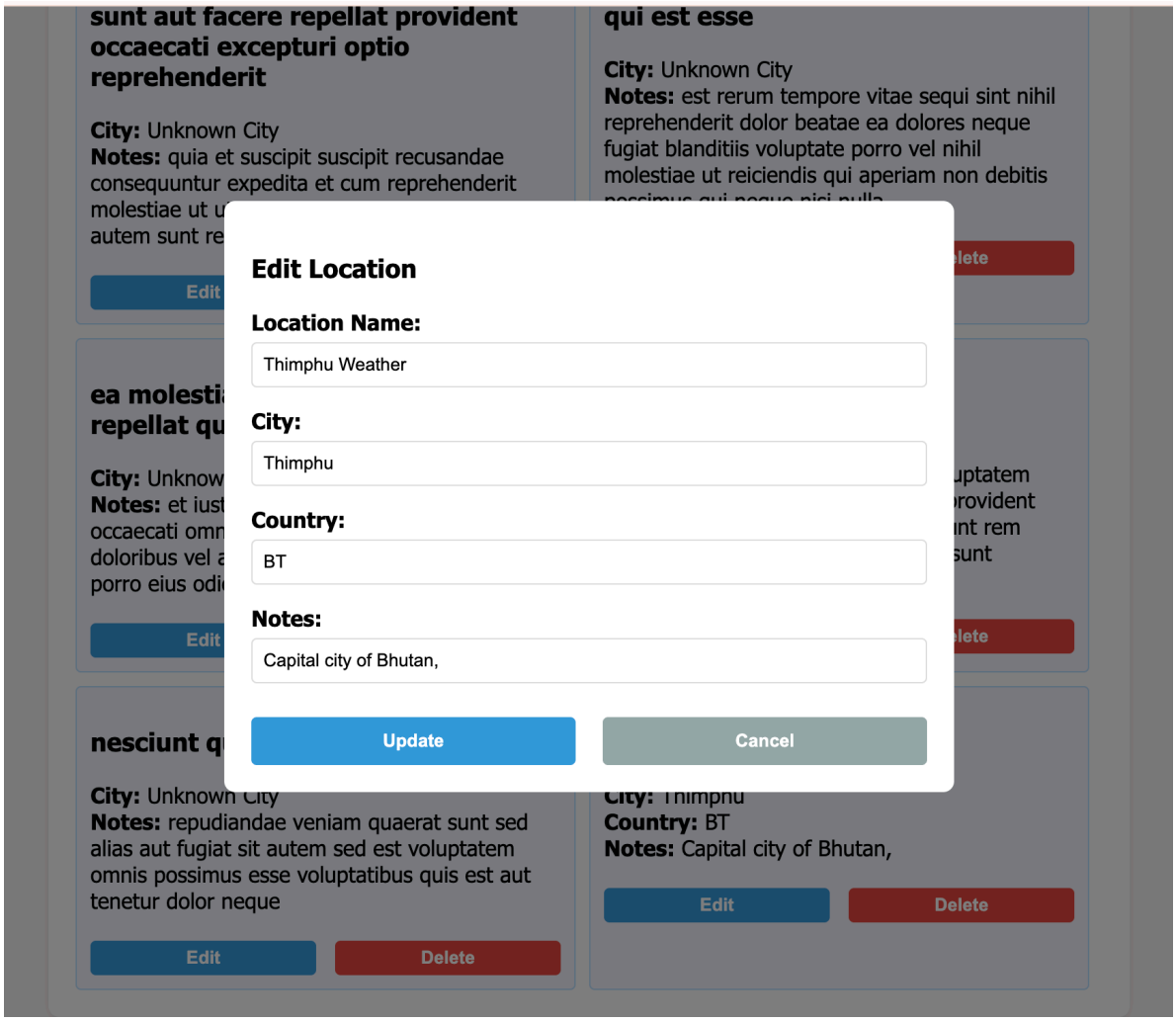


Step 9: POST request tested by saving a new location.

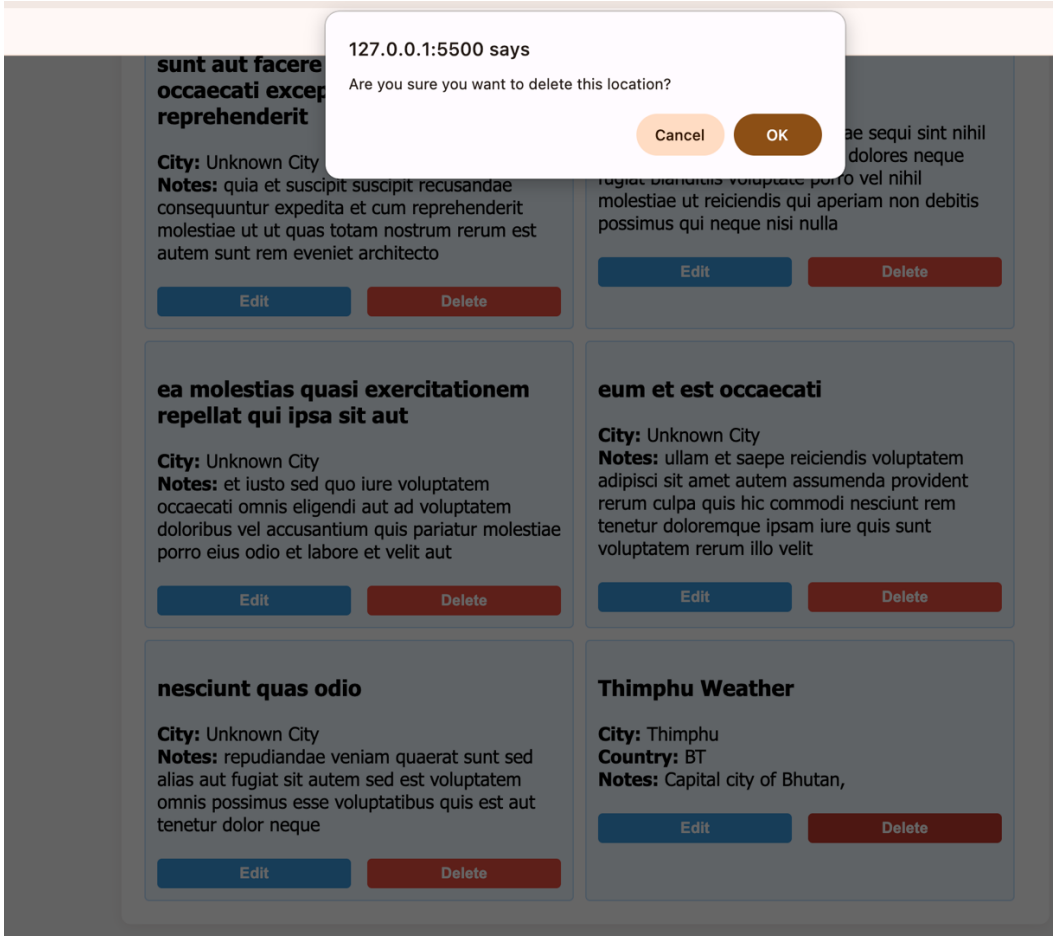




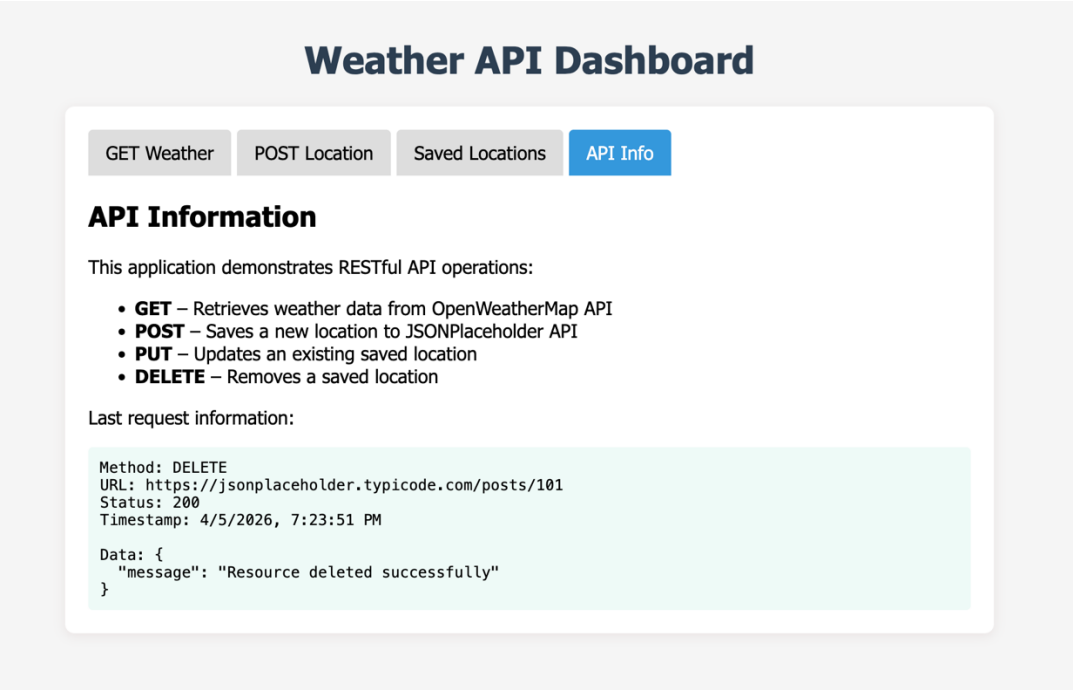
Step 10: PUT request tested by editing a saved location.



Step 11: DELETE request tested by deleting a saved location.

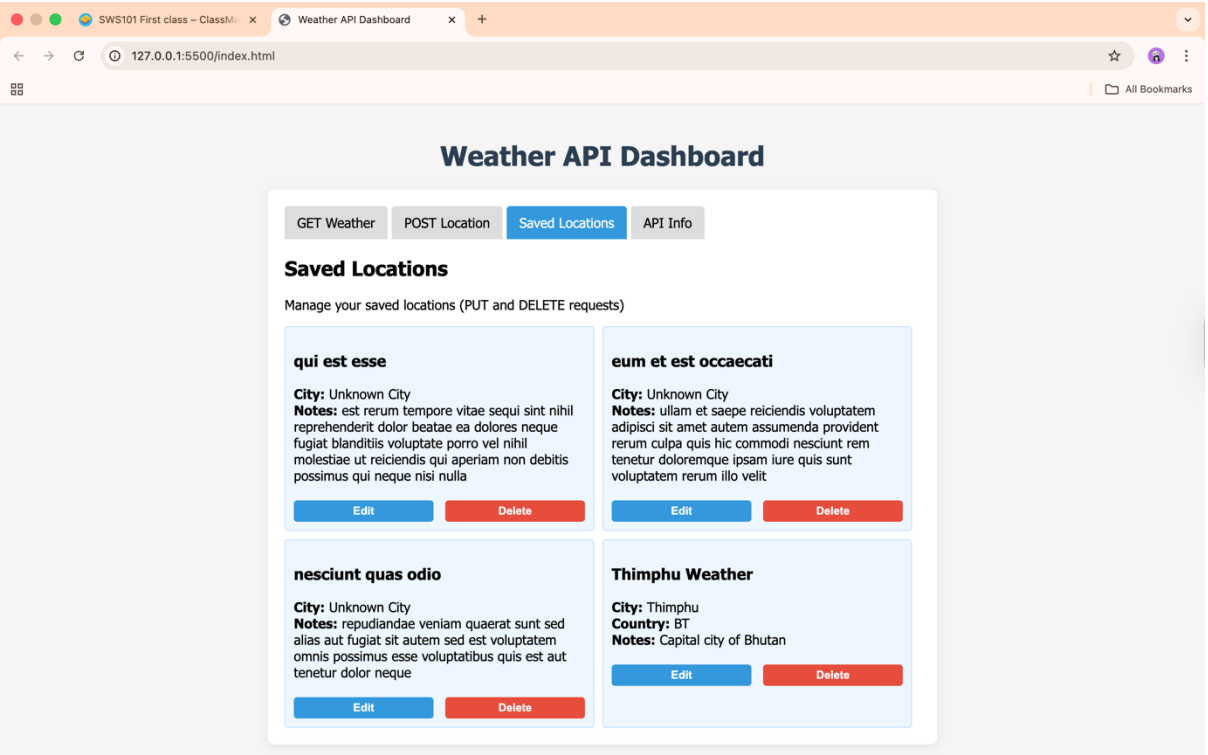


Step 12: API Info tab showing the last request details.



Final Output

This is where I show my talent by showing the final fully workable 4 tabs with functions where it gets the og weather information about any city in this world thought OpenWeatherMap API, the Post tab lets me create new location by filling a details form where is goes to the saved locations with freatures like editing and deleting buttons of the locations. And for the API Info where it will showcase the full detailed information about the recent request activity made having these such as method, URL, status code and the JSON data which come from the server where all the HTTP methods such as GET,POST,PUT and DELETE are working perfectly in the final outcome.



CONCLUSION

Finally, after a lot of errors and syntax I was finally able to successfully make a workable weather API dashboard with 4 tabs that has the ability to retrieve real weather information or data from any city if the world using CSS and JavaScrpit which will use real HTTP requests . This project is divided into 2 parts where in Part 1 I set up the project folders, write the structure for the HTML .and codes for the JavsScript . In Part 2 I made a account in OpenWeatherMap and had to connect it to the fronend by the use of API keys along with proper testing of all the four HTTP methods to ensure its workability.

I faced a lot of problems or issues like the lack of understanding on what the API does since its my first time connecting the frontend to a live server, I had many issues where the API key that I had gotten from the OpenWeatherMap was invalid but silly me it was just taking some to time to get active after signing up for the first time since all of this was my first time working as a website developers that to only frontend and connectivity, another come issue I had was here I could not npm run dev unlike the other practicals so far but later it was easy to just click on the go live icon in the vs code. Gradually I was getting used to it over time with more practice and help from my tutors where I had to carefully use my brains to analyse each error messeges and make the required edits or corrections.

This assignment has given me the chance to build a strong foundation by understanding of REST APIs working on real-live applications in new era of web application which is my first time dealing with these APIs where I found that the HTTP requests work together with the server to fetch API in JavaScript.This new knowledge will definitely be useful down the line in my career path or soon to come assignments, and I will be looking forward to it in the future.

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